Value sensitive Design & its applications to AI

Ethics and Bias in AI : Lecture 4

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Outline

- Value sensitive design
  - Recap
  - Stakeholders
  - Values & Value tensions
    - Designer values
      - What I won’t build
    - Cultural variation
  - Application of VSD to AI/ML
  - Data Statements
    - And other documentation approaches for making values visible

Lecture overview

- Values & technology
- Value Sensitive Design (VSD)
  - Introduce the VSD framework & tripartite structure
  - Stakeholders
  - Value tensions
- Application of VSD to AI/ML

Value Sensitive Design

Values

“a value refers to what a person or group of people consider important in life. In this sense, people find many things of value, both lofty and mundane: their children, friendship, morning tea, education, art, a walk in the woods, nice manners, good science, a wise leader, clean air.”

Friedman, Khan & Borning (2009)

Value Sensitive Design

Value-Sensitive Design (VSD) acknowledges that technology design bears “directly and systematically on the realization, or suppression, of particular configurations of social, ethical, and political values”

(Flanagan et al. 2008).

VSD: Interactional Stance

Friedman & Hendry (2019):

- Design matters. Values can be embodied, at least to some extent, within the features of a tool or technology.
- For example:
  - Electronic document readers, can make text accessible to both blind and sighted people
  - Gender information in a form, free-text vs binary options and radio buttons – more or less inclusive
  - Multiplayer online games can be competitive or cooperative- reflecting or resisting the values of certain groups of people.

Value Sensitive Design

- Human-technology relationship as a dynamic process, working across relatively short and long periods of time.
- We shape technologies, and technologies shape us.
- ‘Technology is neither good nor bad; nor is it neutral’ (Kranzberg, 1986).
- Values can be embodied, at least to some extent, within the features of a tool or technology.
- Unintended uses / non-use by various stakeholder groups.

Value Sensitive Design: an interactional theory

- Values are viewed neither as inscribed into technology nor as simply transmitted by social forces
- A technology's actual use depends on the goals of the people interacting with it.
  - E.g. screwdriver can turn screws but it can also be used as a poker, but functions poorly as a ladle
- Therefore, need to consider potential use cases, but may not be able to predict all

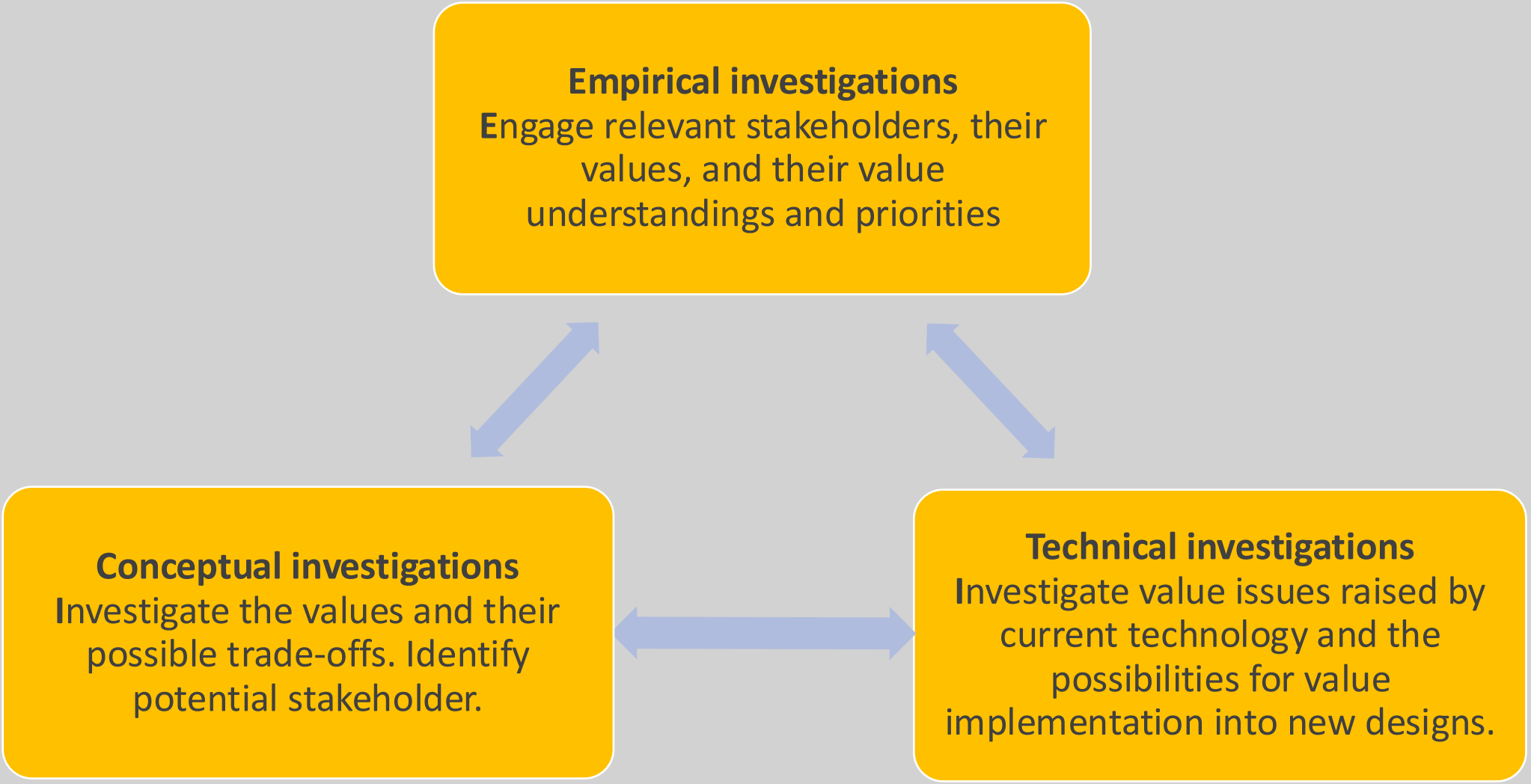
VSD: Interactional Stance

Acknowledges designers do not have complete control…

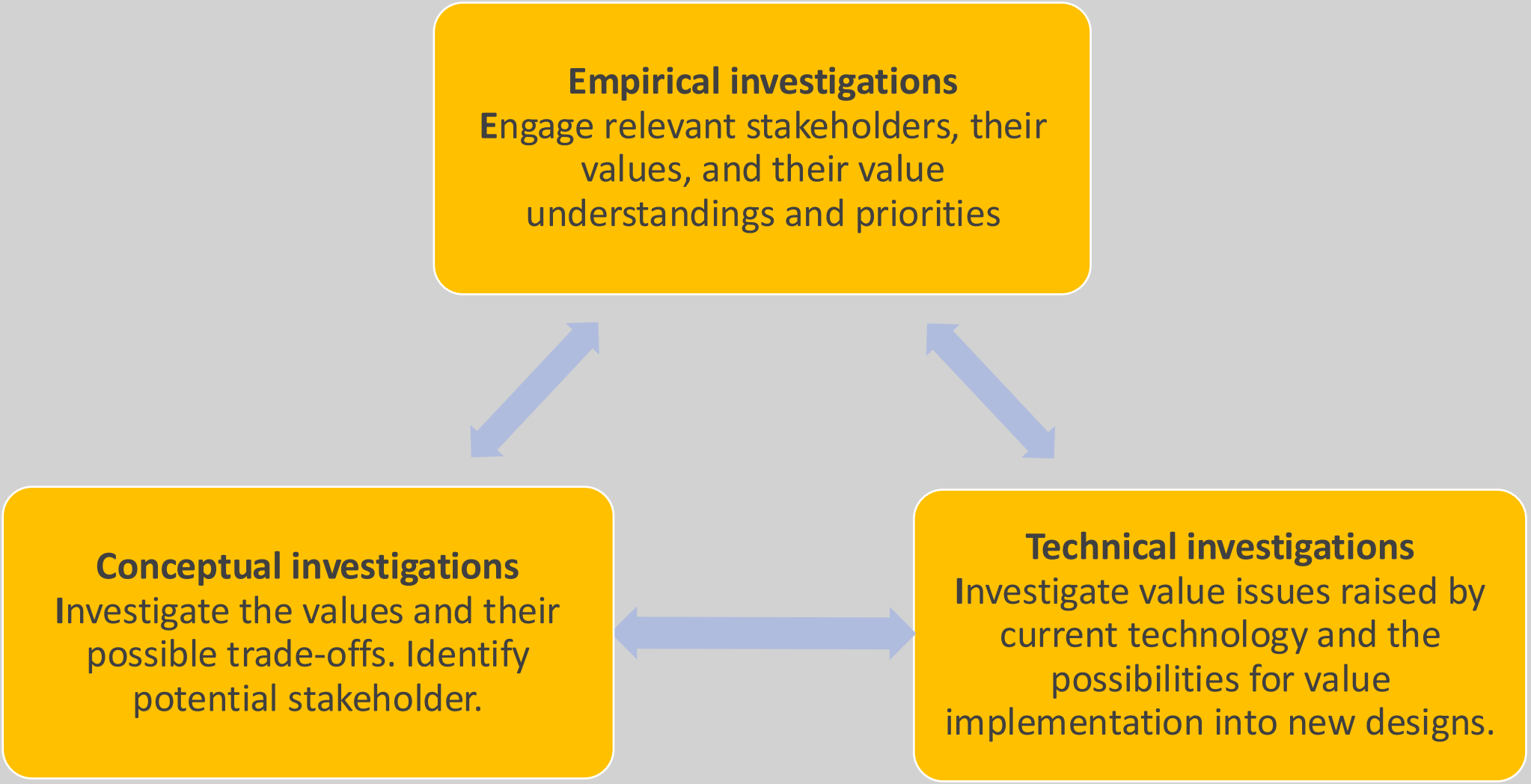
- Different stakeholder groups might use & appropriate a technology according to *their* values, but another group differently, leading to different ends.
- Some stakeholders may decide not to use a technology, perhaps to support a principled position or in keeping with a general disposition.
- During the design phase of a VSD project, after discovery and analysis, the design team might similarly decide not to build a new technology or not to intervene.
- Anticipated and unanticipated effects that arise when placing a technology in different settings / different groups of stakeholders that adopt the technology in different ways & different effects.

  →Design team needs to be sensitive to unintended consequences.

VSD: Tripartite Structure



VSD: Tripartite Structure



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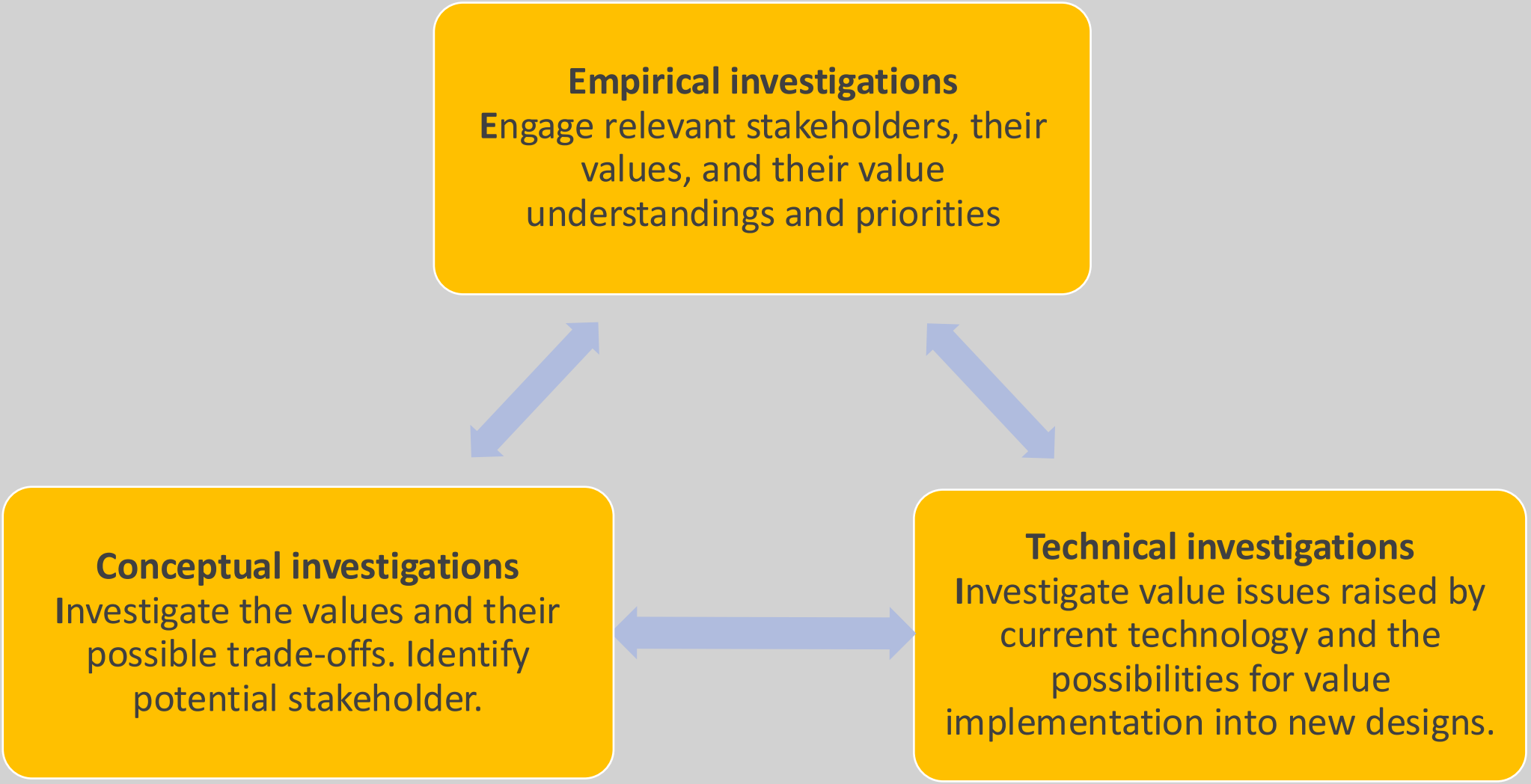

Empirical Investigation Example

- To understand how users feel about privacy, the team conducts empirical research..

For example…

- Surveys: Ask users about their concerns regarding voice data collection.
- Interviews: Speak with families, security experts, and privacy advocates.
- Observational Studies: Monitor how people use voice assistants in real-world settings.
- Analysis of usage data, review data of existing products

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Value Sensitive Design: Key Principles

- Technologies are not neutral. Values can be expressed and embedded in technology.
- Values and choices made by some will have real effects (may not be obvious) on those who are affected (directly/indirectly).
  - Digital technologies (e.g. search engines, content moderation) have real impact on how people act and think.
- Conscious and explicit thinking about the values that are embedded in systems is morally significant.

Aim: to make the particular values at play explicit & evaluate how their implementations works out in practice & adjust accordingly

Value sensitive design: Process

1. Start with a value, technology or context of use
2. Identify direct and indirect stakeholders
3. Identify benefits and harms for each Stakeholder group
4. Map benefits and harms onto corresponding values
5. Conduct an investigation of key values
6. Identify potential value conflicts
7. Integrate value considerations into one's organizational structure

Friedman, Kahn & Borning (2009)

Stakeholders

Values

- Certain values are universally held others may be culturally specific
- How such values play out in a particular culture at a particular point in time can vary considerably
- From the abstract to the concrete
- I.e. Need to carefully unpack what these values mean in practice

Whose values? Who are the stakeholders?

- **Stakeholders:** Those who are or will be significantly implicated by the technology.
- **Stakeholders** can be **people, groups, neighborhoods, communities, organisations, institutions, or societies**.

WHY THESE STAKEHOLDERS?

- VSD asks designers to seek out a robust set of stakeholders groups and to legitimate those stakeholders who have a good deal at stake - that is, to provide an analytical or empirical rationale for their inclusion in the design process.

Stakeholders

Direct stakeholders: Individuals or organizations who interact with a system and its outputs. Often who commissioned it.

Indirect stakeholders: All other parties who are affected by the use of the system (i.e those who don't interact with the system but still impacted by it).

→ **Roles not individuals:** Defining stakeholders in terms of roles positions designers to understand how the same 'entity' could hold multiple relationships to the same technology

VSD: stakeholder identification

To focus the analysis, two overarching stakeholder categories are employed:

(1) direct stakeholders, those who interact directly with the technology

(2) indirect stakeholders, those who do not directly interact with the technology but may nonetheless be affected.

→ May be less clear in AI system design

→ Stakeholders shift roles in different situations

VSD: stakeholder identification

“Depending on the technology, it may be possible to readily identify most, if not all, of the direct stakeholders. Indirect effects can be widespread and diffuse; accordingly, one challenge is to identify those indirect stakeholders who might be significantly impacted, either positively or negatively, by the technology.”

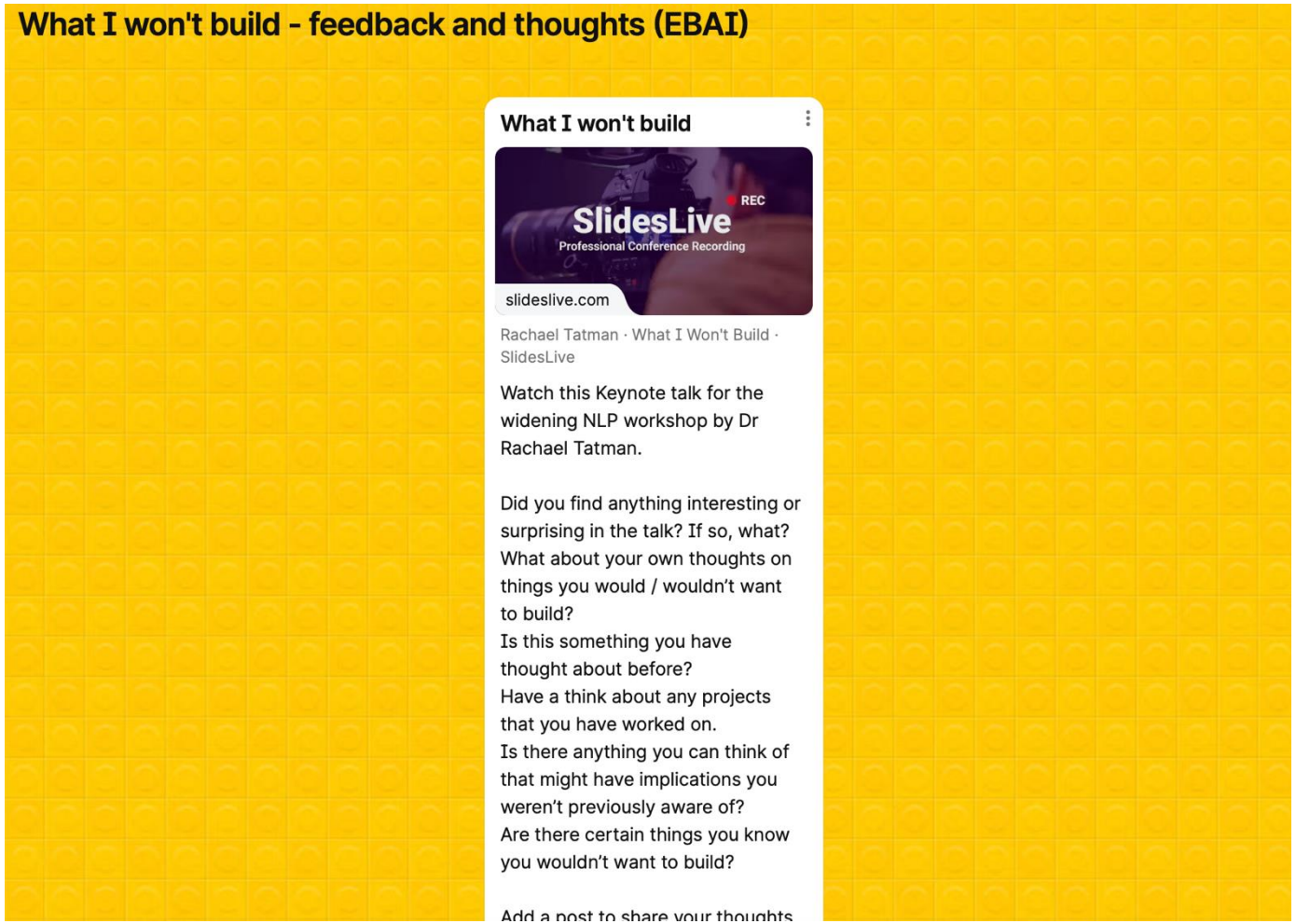
(Friedman & Hendry 2010, p. 65)

Stakeholder values: the role of Designer values

EXPLICITLY SUPPORTED PROJECT VALUES, DESIGNER VALUES AND STAKEHOLDER VALUES:

- Designers bring their own personal and professional values to their work. These are referred to as designer values. In the case of tensions between designer values, project values and stakeholder values, these tensions need to be mitigated = value workshop for both designers & stakeholders

Your values as designers and developers



Designer values & manifestos

Also relevant:

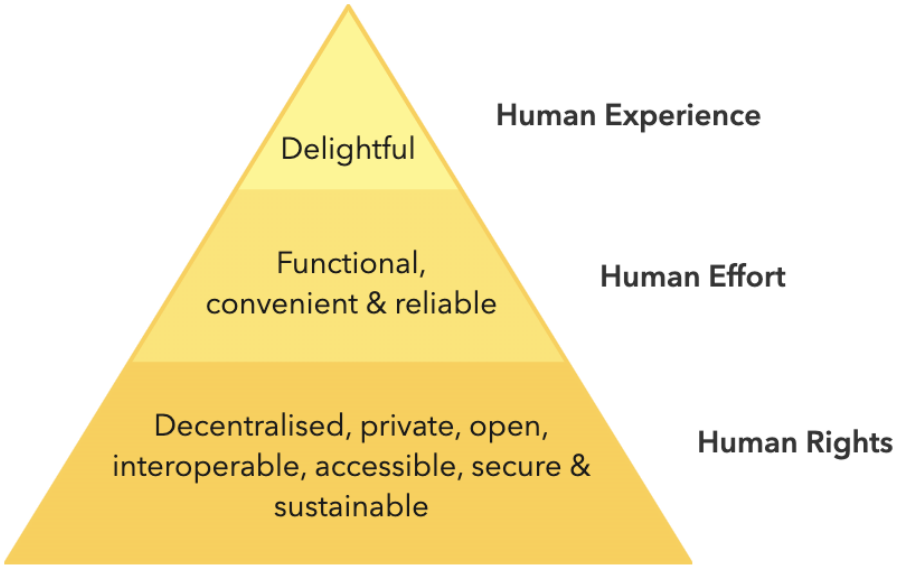
- Project values
- Designer values

[A Manifesto for Decolonising Design]

[Image source: https://designmanifestos.files.wordpress.com/2022/03/a-to-zmanifesto_dissentbydesign.pdf]

Ethical Design

Respect



Human Rights

Technology that respects human rights is decentralised, peer-to-peer, zero-knowledge, end-to-end encrypted, free and open source, interoperable, accessible, and sustainable.

It respects and protects your civil liberties, reduces inequality, and benefits democracy.

Human Effort

Technology that respects human effort is functional, convenient, and reliable.

It is thoughtful and accommodating; not arrogant or demanding. It understands that you might be distracted or differently-abled. It respects the limited time you have on this planet.

Human Experience

Technology that respects human experience is beautiful, magical, and delightful.

It just works. It's intuitive. It's invisible. It recedes into the background of your life. It gives you joy. It empowers you with superpowers. It puts a smile on your face and makes your life better.

Ethical Design Manifesto: https://ind.ie/ethical-design/

Value analysis

- Value analysis should explore the:
  - explicitly supported **project values,**
  - **designer values**
  - **stakeholder values.**
- N.B. Designers bring their own personal and professional values to their work. These are referred to as ’designer values’.

Value tensions

There may be tensions between designer values, project values and different stakeholder values

These tensions should be acknowledged and mitigated (wherever possible)

- Should some values be priortised over others? On what basis?
- One approach to resolve tensions: hold a value workshop for designers & stakeholders.

From abstract values to concrete

van den Hoven, J. (2017)

- Value discourse should not be left on an abstract level
- needs to be operationalized or ‘functionally decomposed’.
- functional decomposition leads to a level of detail that points to specific design features of the system, the ‘moral specs’.

→ This requires engineers to be value-focused in their thinking and capable of articulating the values at play with different stakeholders

VSD: Identifying values

Step 1: Initial project values

- Identify and formulate the preliminary values that the project should rest upon.
- Designer team’s values (e.g. manifestos)
- Based on your assumptions, write down what you think is important to your stakeholders (stakeholder’s initial values),
- Based on yours and your partners (assumed) values, formulate the values that you will base your project upon when you develop the project (initial project values).

VSD: Identifying values

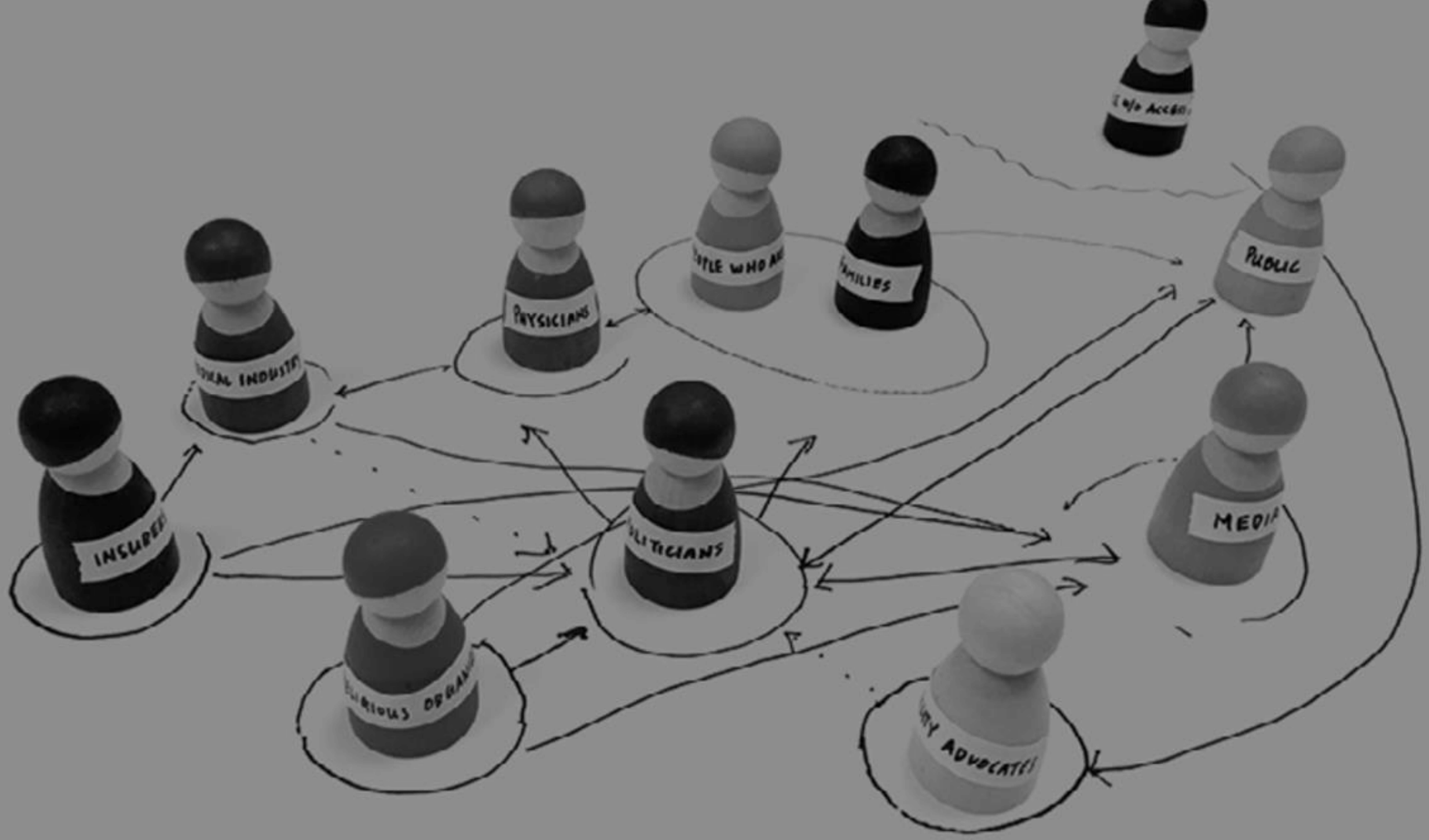
Step 2: Stakeholder's values

- Book a meeting with your stakeholders and discuss what is important to them. Present the document to stakeholders and describe how you think about the chosen values.
- You could invite stakeholders to comment and add extra values that they find important and relate to specific situations. If you meet in a physical location, you can write down all values on a paper sheet. If you meet in a digital meeting, you can use Miro to document your discussion.
- Prioritize values and make a selection of the values that you find most important.

VSD: Mapping of stakeholder’s values

Step 3: Value negotiations

- Write down which values that your stakeholders pointed towards as important
- Write down arguments for why these values were important based on what the stakeholders answered
- Indicate how high each of the values are prioritized (1 – highest, 5 – lowest).
- If possible, suggest how the value can be integrated with the other values.
- This could then be presented back stakeholders and ask them if it corresponds with how they see things.



Mapping stakeholder values - Image source: Friedman & Hendry (2019)

VSD: empirical investigations

Aim: to identify what values are relevant to specific stakeholders, go beyond designer assumptions

- Meetings, interviews, focus groups, surveys

- Ethnographic studies
- Data analysis
- Citizen juries / expert panels (what is an expert)
- Speculative design
- Co-design, participatory design methods

From abstract values to concrete

- Empirical investigations

  E.g. …

  - Observing use of technology (or similar technologies) in situ
  - Engagement with stakeholders (interviews, focus groups, surveys)
  - Experimental investigations
  - Data analysis

Applying Value Sensitive Design to the design of AI systems

Applications of Value Sensitive Design

Project source	Technology domain	Order of investigation (1=first, 2=second, 3=last)			
		Conceptual	Empirical	Technical	Reported iterations
Azenkot et al. (2011)	Mobility facilitation for blind and deaf-blind persons	1	2	3	Yes
Bleumers et al. (2015)	Child–parent play	1	2	3	No
Boyd et al. (2016)	Computer mediated work	1	2	3	No
Brush and Borning (2005)	Daily status messages	1	3	2	No
Cummings (2006)	Command and control supervisory interface	1	3	2	No
Czeskis et al. (2010)	Parental monitoring system	1	3	2	No
Dahl and Holbø (2012)	Sensor-based assistive technology	1	3	2	No
Davis (2008)	Household indicators	1	3	2	No
Epstein et al. (2013)	Sharing of sensed physical activity	1	2	3	No
Friedman et al. (2002)	Informed consent, cookie management	1	2	3	Yes
Kuznetsov (2006)	Contribution motivation	2	1	3	Yes
Miller et al. (2007)	Groupware system for knowledge sharing	1	2	3	Yes
Rector et al. (2015)	Exercise enhancement for the blind	1	2	3	No
Schwartzman and Borning (2007)	Web-based interface for visualizing and browsing indicator results	1	2	3	No
Walton and DeRenzi (2009)	Health information system	1	3	2	No
Watkins et al. (2013)	Transit information tool	1	2	3	No
Xu et al. (2012)	Privacy enhancing tool	1	3	2	No

From: Winkler, T., Spiekermann, S. Twenty years of value sensitive design: a review of methodological practices in VSD projects. *Ethics Inf Technol* **23**, 17–21 (2021). https://doi.org/10.1007/s10676-018-9476-2

Machine learning and VSD

- Not as simple as building and deploying a constrained product….
- It may be not at all clear (to humans) how an AI system has learned certain things.
- The inherent opacity of AI systems requires paying attention to values such as transparency, explainability, and accountability. (Umbrello & van de Poel, 2021)
- Adaptive systems – and then no longer embody the values embedded in them by VSD designers.
- Use-cases may emerge, adapt, shift

AI/ML and Value Sensitive Design

- Traditionally, VSD considered that value implications of a technology should be identified, prior to creating the technology (Friedman et al. 2006).
- Value implications should then be addressed in implementations of the technology to mitigate negative impacts.
- Uncertainties can be resolved by increasing knowledge and understanding from investigations.

→However, for modern technologies, it is often more complex and protracted.

→Therefore, VSD investigations should extend to the entire life cycle rather than just the design phase (de Reuven, 2020; Umbrello & van de Poel, 2021)

Refining VSD to work for the complexities of AI/ML projects… is ongoing

‘Although there is increasing interest in ethics in ~~NLP~~ [AI] there remains the open and urgent question of how we integrate ethical considerations into the everyday practice of our field. This question has no simple answer, but rather will require a constellation of multi-faceted solutions.’

Bender & Friedman, 2018

Bender & Friedman. (2018). Data Statements for Natural Language Processing: Toward Mitigating System Bias and Enabling Better Science. https://doi.org/10.1162/tacl_a_00041

Data Statements - Bender & Friedman (2018)

- A data documentation practice that draws on Value Sensitive Design
- Documentation as method for surfacing and reflecting on values
- Contributes a new professional practice (data statements)
- Aims to bring about improvements in engineering and scientific outcomes while *also* enabling more ethically responsive NLP technology.

Bender & Friedman. (2018). Data Statements for Natural Language Processing: Toward Mitigating System Bias and Enabling Better Science. https://doi.org/10.1162/tacl_a_00041

Data Statement Schema

- Curation Rationale: A statement of the rationale behind the curation of segments in the dataset.
- Language Variety
- Speaker Demographics: Since linguistic variation is found to depend on speaker demographics, authors suggest including: "age, gender, race/ethnicity, native language, socioeconomic status, number of different speakers represent, presence of disordered speech" .
- Annotator Demographics: Since annotator background influences their experience with language, authors suggest including: "age, gender, race/ethnicity, native language, socioeconomic status, training in linguistics/other relevant discipline".
- Speech Situation: "Time and place, modality (spoken/signed, written), scripted/edited vs. spontaneous, synchronous vs. asynchronous interaction, intended audience" (pg. 591).
- Text Characteristics: Genre and topic.
- Recording Quality
- Other Relevant Information

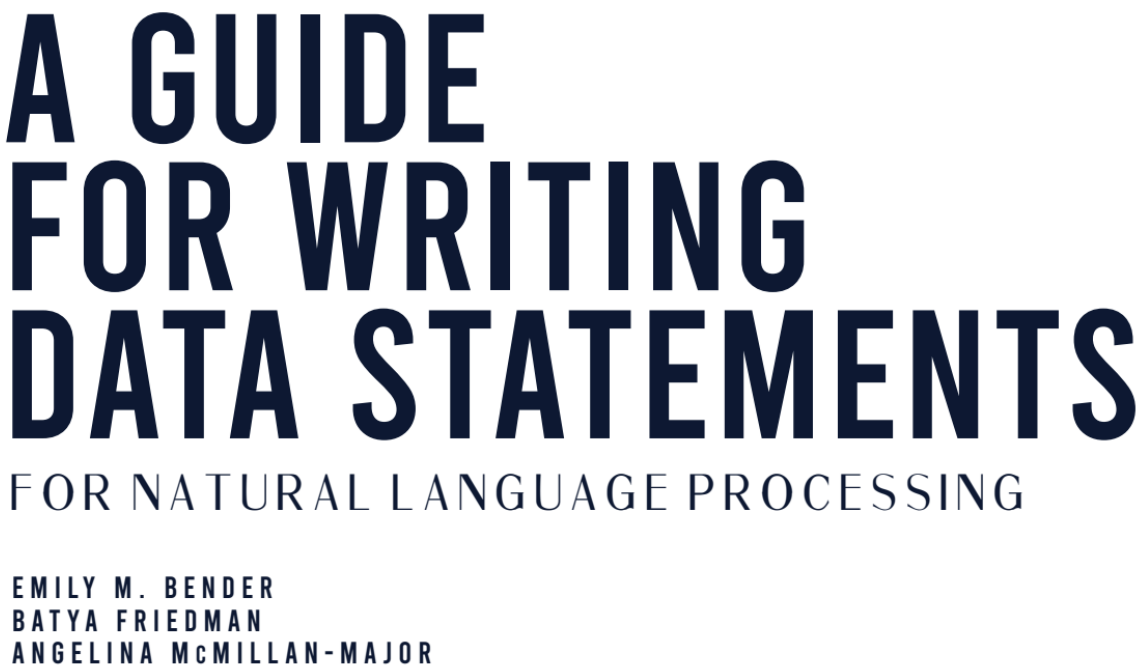
Data Statements - Bender & Friedman (2018)

“[G]ive a characterization of a dataset that **provides context** to allow developers and users to **better understand** how experimental results might generalize, how software might be appropriately deployed, and what biases might be reflected in systems built on the software"

Bender & Friedman. (2018). Data Statements for Natural Language Processing: Toward Mitigating System Bias and Enabling Better Science. https://doi.org/10.1162/tacl_a_00041

Data Statements

- Example Data Statements
- https://ict.fbk.eu/must-she-data-statement/
- An updated guide for writing Data Statements
- https://techpolicylab.uw.edu/wp-content/uploads/2021/10/Data_Statements_Guide_V2.pdf



Datasheets for datasets activity

- Microsoft datasheets for datasets web page: https://www.microsoft.com/en-us/research/project/datasheets-for-datasets/
- Link to academic article: https://www.microsoft.com/en-us/research/publication/datasheets-for-datasets/
- Look at the original version of datasheets template: https://query.prod.cms.rt.microsoft.com/cms/api/am/binary/RE4t8QB

Consider the following:

- What are the benefits of creating documentation for datasets? How might they be useful?
- Can you think of any projects where you might not be able to create a datasheet, why is this?
- Can you foresee any practical challenges or potential objections to integrating datasheets into the development process?
- An updated version of the datasheets template was released in 2022: https://www.microsoft.com/en-us/research/uploads/prod/2022/07/aether-datadoc-082522.pdf
- Does it address any of the issues you flagged? Do any still remain?

Other documentation practices

- Datasheets for Datasets (Gebru et al. 2018; Gebru et al. 2020)
- Dataset Nutrition Labels (Holland et al. 2018; Chmielinski et al. 2020)
- FactSheets (Arnold et al. 2019)
- Model Cards for Model Reporting (Mitchell et al. 2019)
- Nutrition Labels for Data and Models (Stoyanovich and Howe 2019)
- Data Cards (McMillan-Major et al. 2021)

→ developed in a specific research or industry context, with particular authors, users, harms, and use cases in mind.

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